SHORT TITLE OF YOUR PROJECT

Mini Project for DS3294: Data Science Practice

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1 Introduction

Hertzsprung-Russell (HR) diagrams are scatter plots of stellar data, with the y-axis showing increasing luminosity and the x-axis showing colour, spectral class, or decreasing surface temperature. These plots—initially independently formulated by Hertzsprung and Russell—helped astronomers gain a better understanding of stellar evolution.

In this project, we design and implement a reproducible data analysis and visualisation pipeline to explore stellar populations using large astronomical catalogues. Our focus is on constructing and interpreting these HR diagrams while addressing the technical challenges of data volume, measurement uncertainty, and selection effects. By applying quality filters to Gaia DR3 metadata—such as parallax reliability and photometric completeness—we derive the physical quantities necessary to categorize stars and visualize their distribution across different spatial regions.

2 Related Work

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3 Our Contribution

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4 Methods

4.1 Data

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4.2 Name of main method/theory/idea

4.2.1 Data Filtering?

4.2.2 Analysis

The primary approach adopted for the analysis is a comparative population analysis of stellar samples, specifically comparing a near-Earth population ($d < 100$ pc) with a distant field population ($d > 100$ pc)[3]. This method enables a direct observational investigation of selection effects by

examining differences in luminosity functions, effective temperature distributions, Hertzsprung–Russell (HR) diagrams, and spectral class fractions between the two samples.

By comparing these populations, we identify systematic shifts such as the suppression of faint M-dwarfs and the overrepresentation of brighter K and G-type stars in the distant sample as well as the presence of a main sequence in the HR diagram. These differences provide clear empirical evidence of Malmquist bias, arising from the magnitude-limited nature of the survey[1].

A comparative approach is preferred over isochrone fitting because the stellar populations considered here are heterogeneous field stars, spanning a wide range of ages, distances, and evolutionary stages. Isochrone fitting assumes a coeval population with common age and metallicity (star clusters), which is not valid for this dataset. Therefore, a comparison of observed distributions offers a more physically meaningful way to study observational biases in such mixed populations.

4.3 Stellar Parameter Derivations

To construct the HR diagrams, we transform raw Gaia DR3 observations into physical stellar properties using standard geometric and photometric relations [6].

4.3.1 Distance and Absolute Magnitude

The distance d in parsecs is calculated from the parallax ϖ[9]:

$$d = \frac{1000}{\varpi} \tag{1}$$

The G-band absolute magnitude (M_G) is derived by correcting for extinction (A_G) as defined in the Gaia pipeline [6]:

$$M_G = G - A_G + 5\log_{10}(\varpi) - 10 \tag{2}$$

4.4 Absolute Visual Magnitude

The absolute visual magnitude (M_V) is calculated by correcting the apparent V-band magnitude for extinction (A_V) and distance using the parallax (ϖ):

$$M_V = V - A_V + 5\log_{10}(\varpi) - 10 \tag{3}$$

4.4.1 Luminosity

Using the reference solar absolute magnitude $M_{G,\odot} \approx 4.67$ [7], the stellar luminosity L is calculated in solar units ($L_\odot$)[8]:

$$\log_{10}\frac{L}{L_\odot} = 0.4(M_{G,\odot} - M_G) \tag{4}$$

5 Results

5.1 First results: Plots

5.1.1 Luminosity Function

Figure 1 presents the luminosity function for the near-Earth ($d < 100$ pc) and distant field ($d > 100$ pc) stellar populations.

The near-Earth population peaks at faint magnitudes ($M_V \sim 10$–11), corresponding primarily to K and M dwarf stars. In contrast, the distant population peaks at brighter magnitudes ($M_V \sim 4$–5), corresponding to G-type and subgiant stars.

This systematic shift indicates that intrinsically faint stars are well represented in the nearby sample but are largely absent in the distant sample.

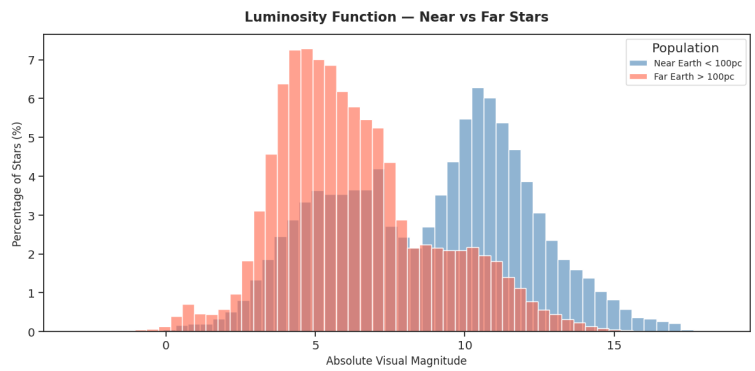

Figure 1: Luminosity function showing the distribution of absolute visual magnitudes for near-Earth ($d < 100$ pc) and distant field ($d > 100$ pc) stellar populations. The shift toward brighter magnitudes in the distant population demonstrates Malmquist bias.

5.1.2 Effective Temperature Distribution

Figure 2 shows the distribution of effective temperatures for both populations.

The near-Earth population peaks at ~ 3000–3500 K, reflecting the dominance of cool M dwarf stars. In contrast, the distant population peaks at higher temperatures (~ 4500–6000 K), corresponding to K and G-type stars, with a noticeable suppression of the cool-star peak.
This difference arises because cool M dwarfs are intrinsically faint and become undetectable beyond ~ 100 pc, leading to a biased temperature distribution in the distant sample.

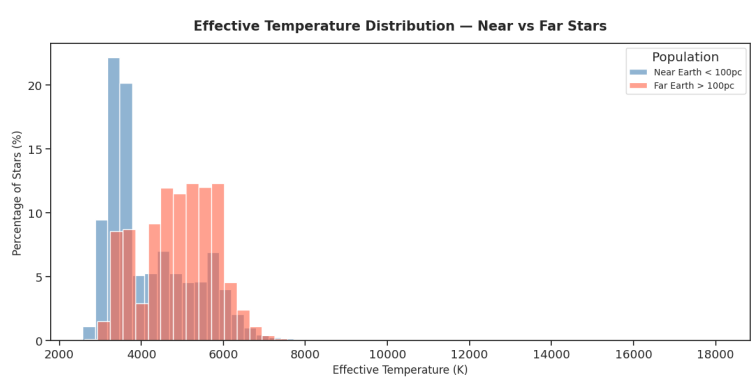

Figure 2: Effective temperature distribution for near-Earth and distant stellar populations. The nearby sample is dominated by cool M dwarfs, while the distant sample is biased toward warmer stars due to observational limits.

5.1.3 HR Diagram and Spectral Distribution

Figure 3 presents the Hertzsprung Russell (HR) diagrams for both populations. In both cases, the main sequence is visible as a diagonal band from hot, luminous stars to cool, faint stars.

However, the near-Earth sample extends to very low luminosities, clearly showing the lower main sequence populated by M dwarfs. In contrast, the distant sample is truncated at the faint end, with the lower main sequence largely absent.

Figure 4 further quantifies this effect through spectral class distributions. M-type stars dominate the nearby sample (approximately 50%) but are significantly underrepresented in the distant sample (approximately 16%). Conversely, brighter K and G-type stars are overrepresented at larger distances.

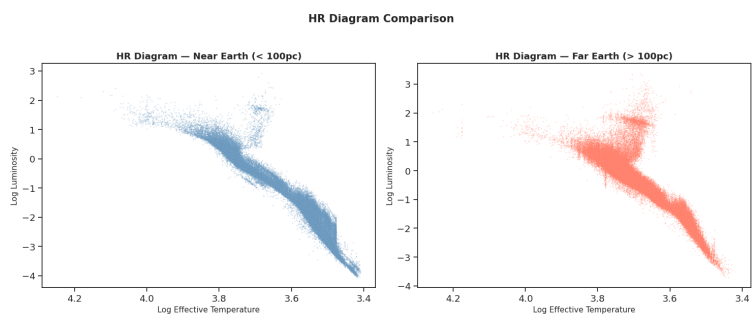

Figure 3: Hertzsprung–Russell diagrams for near-Earth and distant stellar populations. The distant population is truncated at the faint, cool end due to the detection limit.

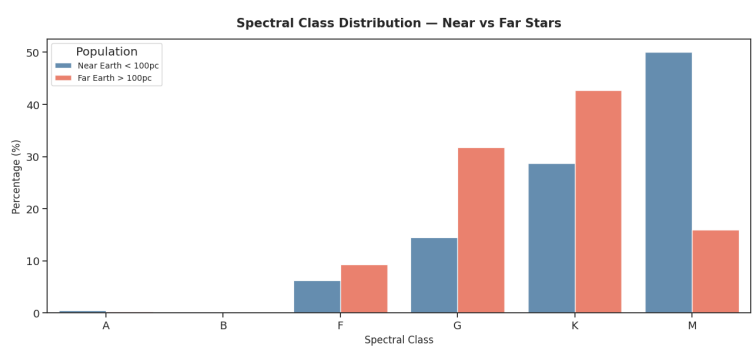

Figure 4: Spectral class distribution for near-Earth and distant populations. M dwarfs dominate the nearby sample but are underrepresented at larger distances, demonstrating Malmquist bias.

5.2 Second results: Dashboard

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5.3 Third results: Use case

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6 Conclusion and Outlook

In this study, we analysed stellar populations using Gaia DR3 data. By comparing a near-Earth sample ($d < 100$ pc) with a distant field sample ($d > 100$ pc), we examined how observational selection effects influence the inferred properties of stellar populations.

The luminosity function revealed a clear shift toward intrinsically brighter stars in the distant sample, highlighting the inability of magnitude-limited surveys to detect faint objects such as M dwarfs at larger distances. This effect was further supported by the effective temperature distribution, where the nearby population is dominated by cool ($\sim$3000 K) M-type stars, while the distant sample shows a bias toward warmer K- and G-type stars.

The Hertzsprung–Russell diagrams provided strong visual evidence of this bias, with the distant population exhibiting a truncated main sequence at the faint, cool end. Similarly, the spectral class distribution quantitatively confirmed the underrepresentation of M dwarfs and the overrepresentation of brighter stellar types in the distant sample.

Overall, our results demonstrate that the apparent differences between nearby and distant stellar populations are not purely physical, but are largely driven by observational limitations. Malmquist bias must therefore be carefully accounted for in any statistical study of stellar populations.

A potential direction for future work is the use of dynamical population synthesis, where the Galactic field stellar population is modeled as a superposition of contributions from star clusters formed at different times and distances. In this framework, field stars are not treated as an independent population but as the combined outcome of cluster formations. Such approaches provide a more physically guided solution by linking field star properties directly to cluster evolution [2].

7 Acknowledgements

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8 Author Contributions

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9 Code Availability

The code for this project is available at: `https://github.com/bedartha/cl-todo`

10 References

References

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[4] L. Fillioux, J. Silva-Rodríguez, et al. *Are foundation models for computer vision good conformal predictors?*, Mar. 2025. arXiv:2412.06082. [cite: 166, 167]

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[9] T. E. Lutz and D. H. Kelker. *On the Use of Trigonometric Parallaxes for the Calibration of Luminosity Systems.* Publications of the Astronomical Society of the Pacific, 85(507):573, 1973.

11 Appendix

11.1 Additional Figures

Additional figures can go here.

11.2 Algorithms

Algorithm 1 An algorithm with caption

Require: $n \geq 0$
Ensure: $y = x^n$

```
y ← 1
X ← x
N ← n
while N ≠ 0 do
    if N is even then
        X ← X × X
        N ← N/2
    else
        y ← y × X
        N ← N − 1
    end if
end while
```

11.3 Code Snippets

```
# Function to check prime numbers
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

# Print primes from 1 to 20
for i in range(1, 21):
    if is_prime(i):
        print(f"{i} is prime")
```